

Materi Kuliah

Pertemuan ke-2 :

Pemrograman Multimedia

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2D dan 3D

Character Controller

Input

```
Unity Script (1 asset reference) | 0 references
5 public class Player : MonoBehaviour
6 {
7     // Update is called once per frame
8     Unity Message | 0 references
9     void Update()
10    {
11        Vector2 inputVector = new Vector2(0, 0);
12
13        if (Input.GetKey(KeyCode.W))
14        {
15            inputVector.y = +1;
16        }
17        if (Input.GetKey(KeyCode.S))
18        {
19            inputVector.y = -1;
20        }
21        if (Input.GetKey(KeyCode.A))
22        {
23            inputVector.x = -1;
24        }
25        if (Input.GetKey(KeyCode.D))
26        {
27            inputVector.x = +1;
28        }
29    }
```

Character Controller

Movement

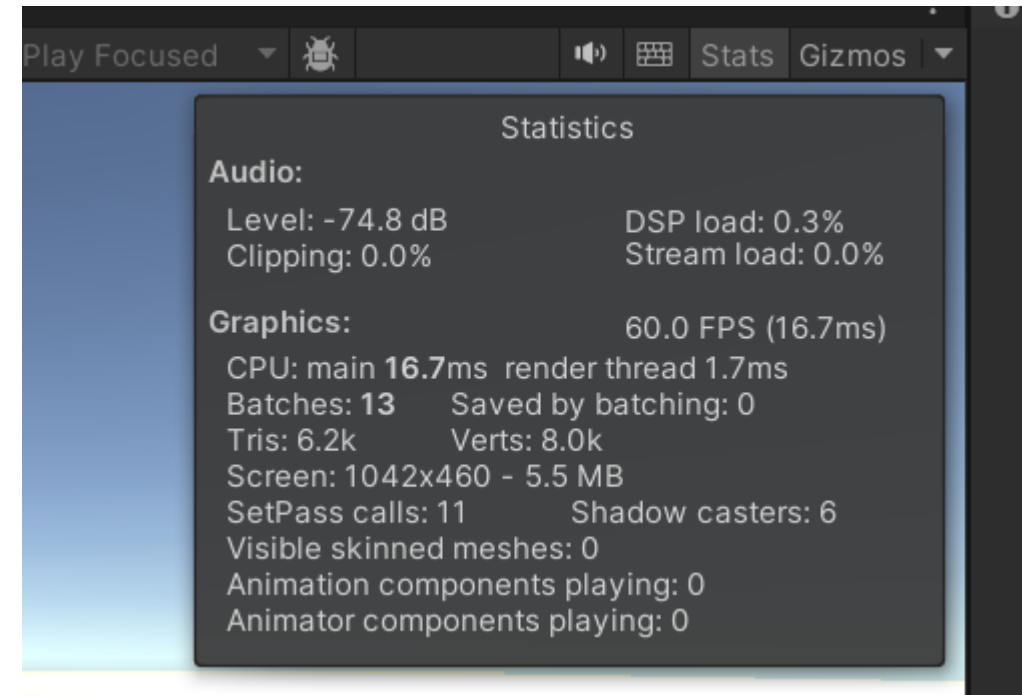
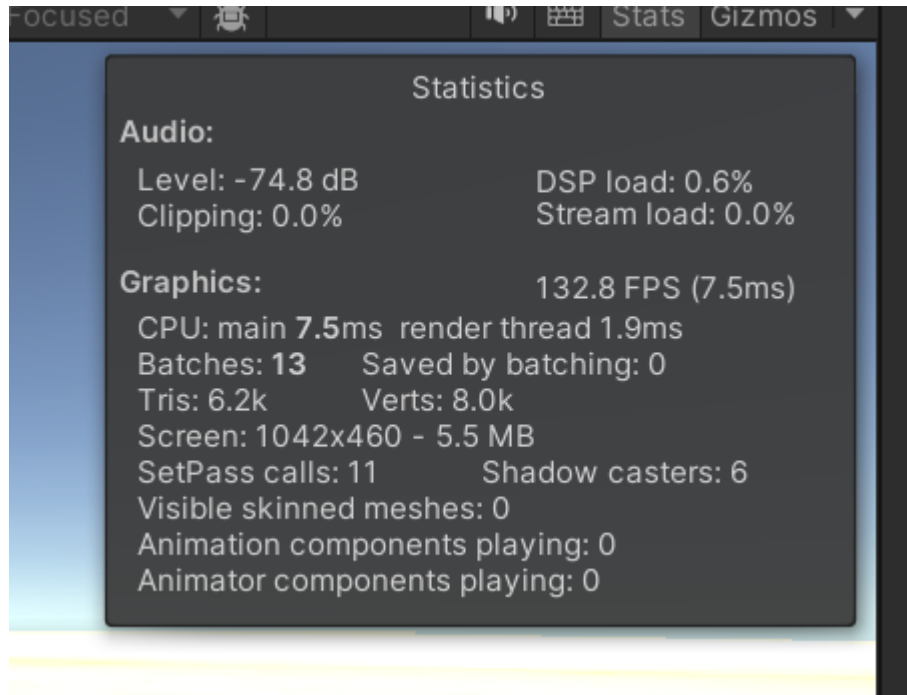
```
32  
33 transform.position += (Vector3)inputVector;  
34  
35
```

Problem: Karakter bergerak ke sumbu x (kanan/kiri) dan sumbu y (atas/bawah). Sementara yang dibutuhkan adalah Kanan/kiri dan depan/belakang

```
34 Vector3 moveDir = new Vector3(inputVector.x, 0f, inputVector.y);  
35 transform.position += moveDir;  
36
```

Character Controller

Speed Configuration



```
Vector3 moveDir = new Vector3(inputVector.x, 0f, inputVector.y);  
transform.position += moveDir * Time.deltaTime;
```

Character Controller

Speed Configuration

What does `Time.deltaTime` do?

That field contains the number of seconds elapsed since the last frame was rendered. So if the game is running at 10 FPS then that field will have 0.1, meaning every frame takes 0.1 seconds to render so 10 frames render in 1 second.

By using it in the calculation you are making sure the character movement is based on time regardless of framerate.

If the game is running at 10 FPS then the character will move 0.1 units per frame so it will move 1 unit in 1 second.

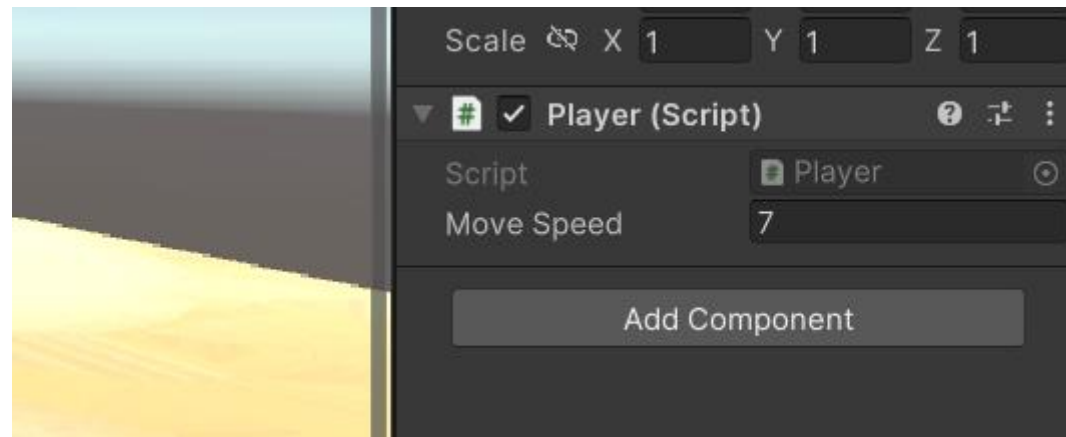
If the game is running at 50 FPS then the character will move 0.02 units per frame so it will move 1 unit in 1 second.

Character Controller

Speed Configuration

```
6 public class Player : MonoBehaviour
7 {
8     public float moveSpeed = 7f;
9 }
```

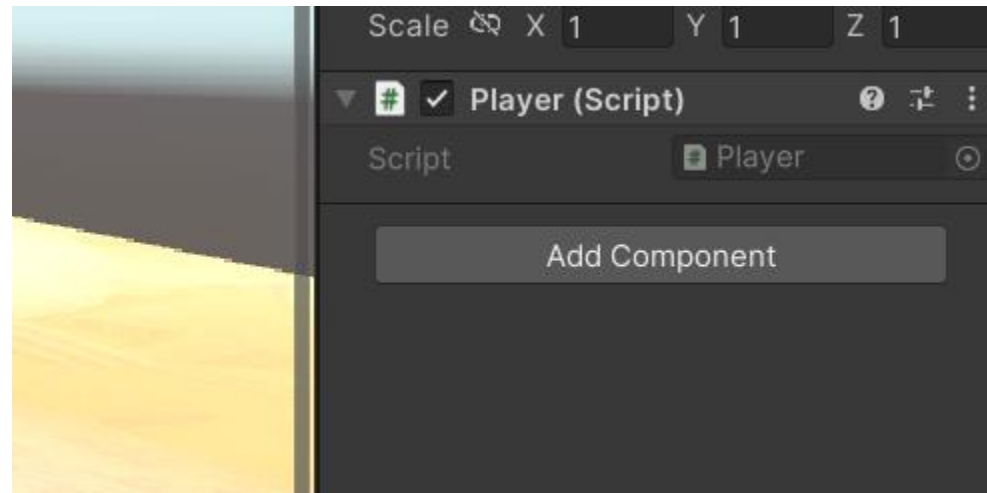
```
35
36 Vector3 moveDir = new Vector3(inputVector.x, 0f, inputVector.y);
37 transform.position += moveDir * moveSpeed * Time.deltaTime;
38
```



Character Controller

Speed Configuration

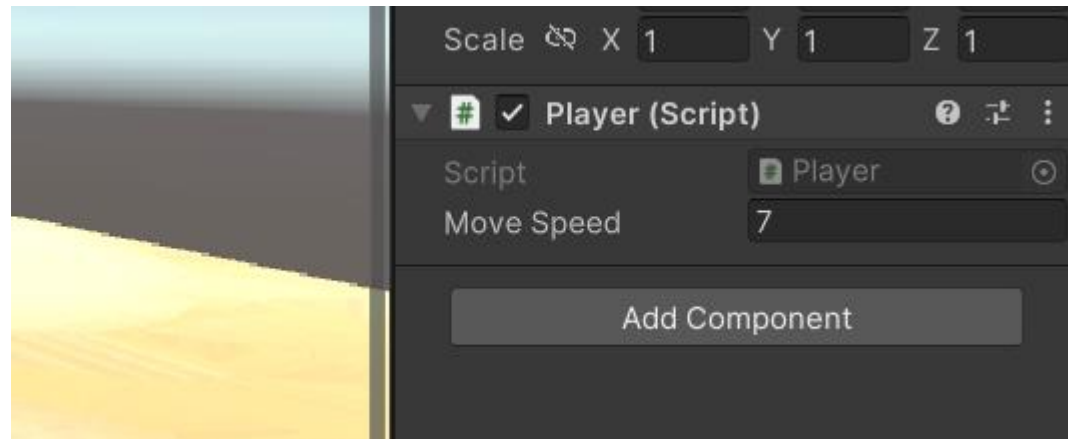
```
6 public class Player : MonoBehaviour
7 {
8     private float moveSpeed = 7f;
9 }
```



Character Controller

Speed Configuration

```
6 public class Player : MonoBehaviour
7 {
8     [SerializeField] private float moveSpeed = 7f;
9 }
```



Character Rotation

Rotating configuration

```
transform.forward = moveDir
```

Bagian depan karakter langsung menyesuaikan arah keypad tanpa ada rotasi

```
transform.forward = Vector3.Slerp(transform.forward, moveDir, Time.deltaTime)
```

Character Rotation

Rotating speed configuration

```
float rotateSpeed = 5f
```

```
transform.forward = Vector3.Slerp(transform.forward, moveDir, Time.deltaTime *  
    rotateSpeed)
```

Source

<https://www.youtube.com/watch?v=AmGSEH7QcDg&t=6162s>



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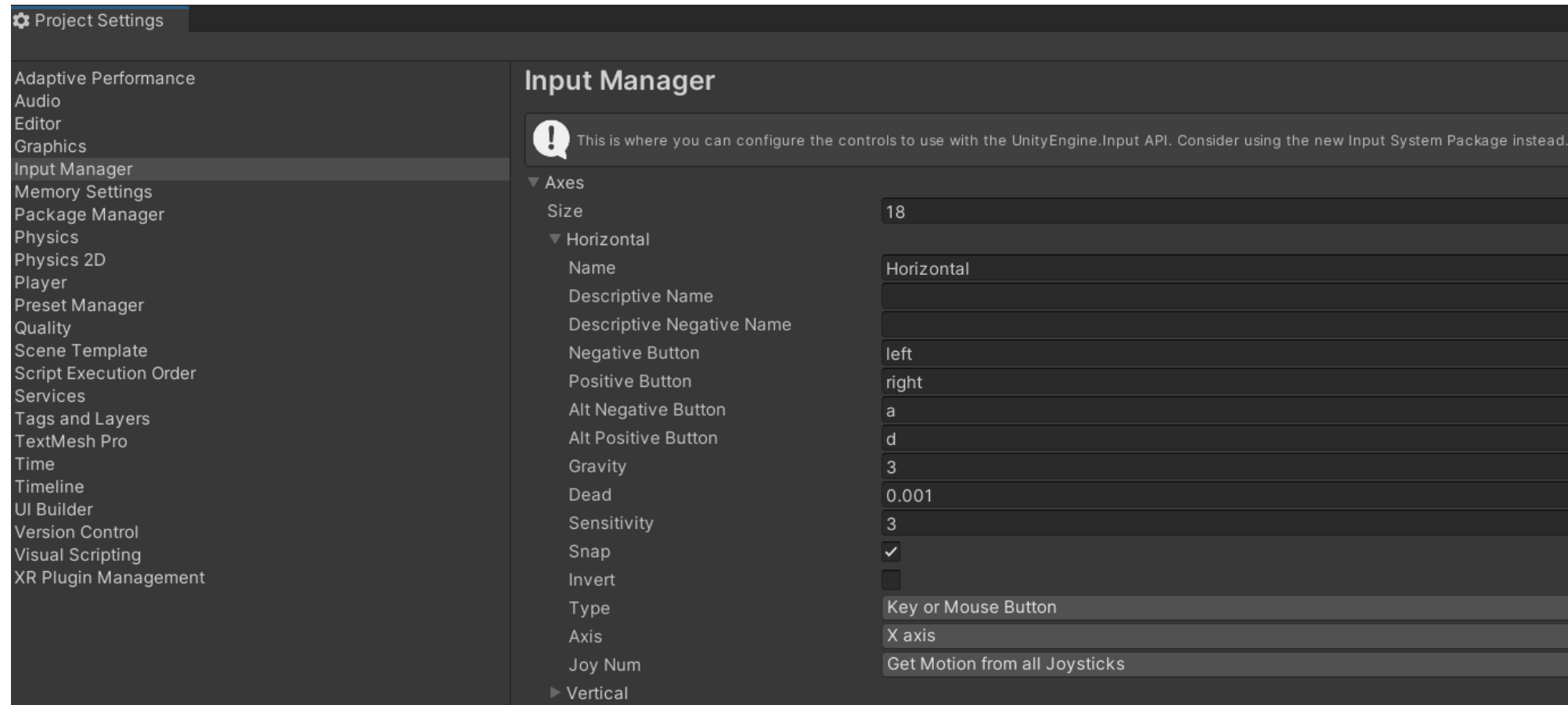


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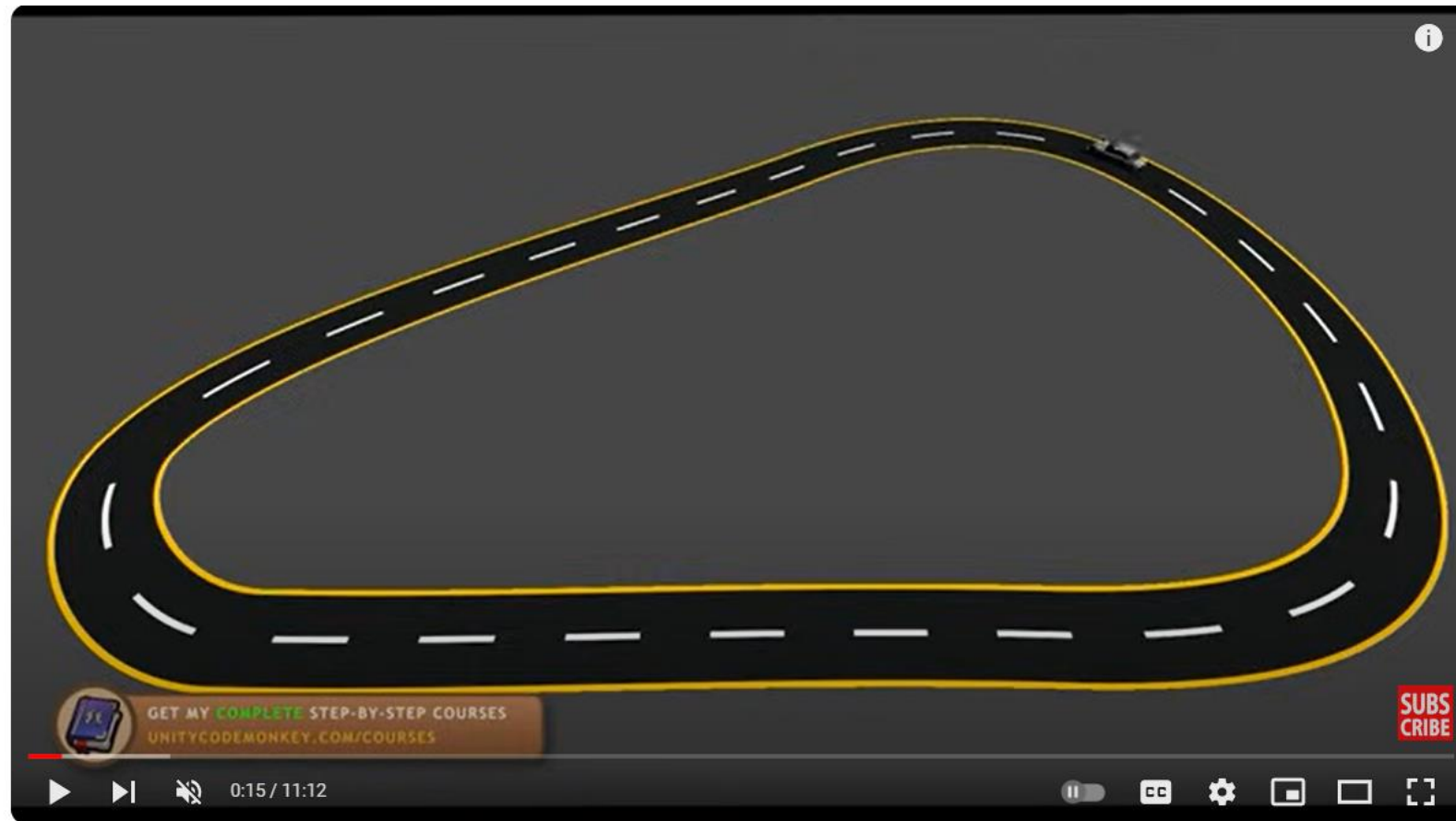


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